



Multi-timescale risk-averse restoration for interdependent water–power networks with joint reconfiguration and diverse uncertainties

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ABSTRACT

The growing interdependencies between water and power systems have increased the risk of cascading disruptions and widespread blackouts. Such interdependencies, together with different operational characteristics and multiple uncertainties, introduce additional complexities to service restoration. To address these issues, this paper proposes a coordinated multi-timescale restoration strategy for interdependent water–power networks (IWPNS). First, we model the IWPNS as network-based with physical mechanisms, incorporating component-wise interdependencies and varying consumer demands. Features comprising pipe damage (water leakage) and storage as well as renewable generations are modelled to better reflect restoration better. Specifically, the joint reconfigurability of water and power networks is first applied for adjustment of topologies and leverages off backup components by coordinated setting of valves and switches. Then, an updated estimation for multiple uncertainties during restoration is utilized, which offers increasing clarity to support better decision-making. These uncertainties arise from renewable generations and water and power demands. A multi-timescale decision framework is developed to capture these effects and tune restoration measures based on response speeds to facilitate efficient and reliable restoration. Finally, the method is implemented by combining robust optimization and risk-averse stochastic programming and applied to a community-scale test system with 25 water nodes and 33 power buses. The proposed method is compared with five conventional methods with numerical results demonstrating the improvements arising from an interdependent restoration, joint reconfigurability, and multi-timescale optimizations.

1. Introduction

The reliable and resilient operation of interdependent water and power networks (IWPNS) is paramount for ensuring the functioning of modern societies via ensuring critical water and energy services [1]. However, IWPNS are susceptible to disruptions caused by natural disasters, equipment failures, and other unforeseen events, leading to significant societal and economic impacts [2]. Therefore, an effective and efficient restoration of IWPNS service in the aftermath of such disruptions is of utmost importance [3]. Given this, more reliable restoration strategies for IWPNS, combining effective management of multiple restoration measures (RMs), and responsiveness to uncertainties is a continuing challenge [4].

In the literature, modelling of IWPNS typically utilize agent-based and network-based models [5]. Agent-based methods treat components in IWPNS as agents and simulate their interactions, while neglecting the detailed network structures. For example, the authors in [6] built

an IWPNS model based on multiple agents, representing thermal plant, wind, solar, hydro generation, and operator agents. In [7], an agent-based model was built for restoring water distribution systems with faulted power supply and limited backup electricity. On the other hand, network models depict the components of IWPNS as nodes and links which allow detailed physical representations of the linkages and interdependencies. An IWPNS model is developed in [8] that includes pumps and tanks in the water system as well as microgrids in the power system. However, these two subsystems are modelled via different methods, leading to limited interdependencies between them. To better consider comprehensive interdependencies, the study in [9] integrally modelled water and power systems by employing network formations. The developed model allows component-wise interdependent linkages though it still neglects key operational details, e.g., voltage and pressure. The authors in [10] proposed a network model for coupled power-water systems using physical mechanisms and

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Nomenclature

Abbreviation

WDN/PDN	Water/Power distribution network
MG/DG	Microgrid/Distributed generator
ESS/SOC	Energy storage system/State of charge
WBPS	Water booster pressure system

Indices and Sets

$\mathcal{J}$	Set of nodes in WDN, indexed by i, j
$\mathcal{L}$	Set of pipelines in WDN, indexed by (i, j)
$\mathcal{I}$	Set of buses in PDN, indexed by i, j
$\mathcal{L}$	Set of branches in PDN, indexed by (i, j)
$\mathcal{T}$	Set of time steps, indexed by t

Parameters

$\Delta_{i,j}^{tr}$	Interdependence for treatment plants
$\Delta_{i,j}^{wb} / \Delta_{i,j}^{nex}$	Interdependence for WBPS/demands
$\Delta_{i,j}^{tk} / \Delta_{i,j}^{pu}$	Interdependence for tanks/pump stations
$c_{i,j}^{leak} / r_{i,j}^{air}$	Pipe leakage/storage coefficient
u_1 / u_2	Indicators for location of leakage
$\mathcal{V}_{i,j} / \Delta T$	Pipe volume and time span
$r_{i,j}^{lim} / r_{i,j}^{vol}$	Maximum rate and volume for air replacement
$(Q_{i,j} / Y_{i,j}^{pu})^{\max}$	Maximum water flow and pump power
$(\alpha / \beta / \gamma)^{hw}$	Constants for Hazen–Williams equation
$L_{i,j} / C_{i,j} / D_{i,j}$	Length, roughness, and diameter of pipe
$\mathcal{E}_{i,j} / \mathcal{H}_{i,j}^{tk}$	Elevation of water node and head rise for tank
$\mathcal{P}_{i,j}^{des} / \mathcal{P}_{i,j}^{\min}$	Design/minimum pressure
$\varepsilon_{i,j}^{dl}$	Binary acceptance for delayed demand
$r_{i,j} / x_{i,j} / S_{i,j}^{\max}$	Resistance/reactance/capacities of branches
$V^{\max} / \min / 0$	Maximum/minimum/reference voltage
$(P/Q)_i^{dg, \max}$	Maximum active (reactive) power for DG
$(P/Q)_i^{dg, \min}$	Minimum active (reactive) power for DG
E_i^{ess}	Capacity of ESS
$SOC_i^{\max} / \min$	Maximum/minimum of SOC
$P_i^c / d, \max$	limits for ESS charging/discharging
$E_{i,j}^{cap} / t_{i,j}^{dvr}$	Stock fuel capacity/arrival time
$r_{i,j}^{tr} / r_{i,j}^{wb}$	Power consumption for treatment/WBPS
ρ / g	Water density and acceleration
η_i^p / η_i^t	Efficiency for tank-used pump and turbine
$\mathcal{V}_i^0 / \mathcal{V}_i^{\max}$	Initial/maximum volume of tank
$\sigma^a / \sigma^b / \sigma^c$	Coefficients for pump equations
$\varpi_{i,j}^{I, \mathcal{J}} / \varepsilon_{i,j}^{I, \mathcal{J}}$	Coupling rates for power/water demands
$\eta_{i,j}^{pu}$	Power efficiency for pump stations
$\omega_{i,j}^w / \omega_{i,j}^p$	Weights of water/power load
$c_{i,j}^w / c_{i,j}^p$	Costs for valve and switch actions
$Q_{i,j}^{tr, \max}$	Maximum water from water treatment
$Y_{i,j}^{pc} / pg, \max$	Energy use and recovery limits for tank pump
M^W / M^P	Big scatter for water/power model
$I_i^{rt} / \mathcal{J}_i^{rt}$	Binary state for potential root bus/node
α / ρ	confidence level of scenarios and weight for risk measure

Decision variables

$Q_{i,j,t} / Q_{i,j,t}^l$	Water flow and leakage of pipe
$\mathcal{H}_{i,t} / \mathcal{P}_{i,t}$	Water head/pressure
$Q_{i,j,t}^{air} / \mathcal{V}_{i,j,t}^{air}$	Change and accumulation of air in pipe
$Q_{i,t}^{tr} / Y_{i,t}^{tr}$	Water production and power use of treatment
$Q_{i,t}^c / Q_{i,t}^d$	Water inflow/outflow of tank
$\mathcal{W}_{i,t}^r / \mathcal{V}_{i,t}^{tk}$	Water load restoration and storage in tank
$\mathcal{H}_{i,j,t}^{loss(air)}$	Head loss in pipes (with air replacement)
$\mathcal{H}_{i,t}^{pu} / Y_{i,t}^{pu}$	Head gain and power use of pump station
$\Delta \mathcal{W}_{i,t}^{ud} / \mathcal{W}_{i,t}^{buf}$	Unmet water demand and buffer for record
$P_{i,t}^{dg} / Q_{i,t}^{dg}$	Active/reactive power generation of DG
$P_{i,t}^c / P_{i,t}^d$	Charge/discharge power of ESS
$P_{i,t}^r / Q_{i,t}^r$	Active and reactive power load restoration
$P_{i,j,t} / Q_{i,j,t}$	Active/reactive power flow
$Y_{i,t}^{wb/pg} / pg$	Power use of WBPS and use/recovery of tanks
$Q_{i,t}^w / Q_{i,t}^{pv}$	Reactive wind and solar generation
$V_{i,t} / SOC_{i,t}$	Voltage magnitude and SOC for ESS
$E_i^{ex} / E_{i,t}^{dg}$	Fuel pre-allocation and remaining fuel for DG
$\mu_j^W / \lambda_{i,j}^W$	Binary root node/pipeline connection in WDN
$\mu_i^P / \lambda_{i,j}^P$	Binary root bus/branch connection in PDN
$\vartheta_{i,j} / \theta_{i,j}$	Fictitious flow for WDN and PDN

Random variables

$\tilde{\mathcal{W}}_{j,t}$	Uncertain water loads
$\tilde{P}_{i,t} / \tilde{Q}_{i,t}$	Uncertain activate and reactive power loads
$\tilde{P}_{i,t}^w / \tilde{P}_{i,t}^{pv}$	Uncertain wind and solar generation

component-level interdependencies. However, it is difficult to scale up spatially or temporally due to computation complexity. For all the discussed Refs. [6–10], the models only dealt with the stable operation of the IWP, some features exhibited under emergencies are not captured, such as pipe breakage and leakage, water stored in pipes, and delayable water demand. Lastly, couplings between water and power demands should also be considered [11]. It is noted that other methods such as system dynamic models [12] and economic models [13], also model the network and component features of IWPNs but in an indirect way. As such they are less used for modelling IWPNs compared to agent-based and network-based models.

Traditionally, restoration efforts for IWPNs are carried out individually by water and power utilities. For example, the research detailed in [14] proposed a real-time pump scheduling method to restore water supply after disruptions. In this method, the power outage is regarded as constant instead of variable. A pumped-hydro storage (PHS) system combined with wind power is utilized in [15] to facilitate a self-healing power grid. However, the impacts on the water system that connects to the PHS are neglected. In most cases, disruptions do not damage only the water or electricity system, but both [16]. For example, a cold wave in Texas U.S. in February 2021 caused the power grid's failure leading to widespread outages and extensive damage to water infrastructure with pipes freezing and bursting. Over 4.5 million homes and businesses were affected [17]. A further example is the Typhoon Yagi which hit China's Hainan on 6th September 2024 and left more than 45 neighbourhoods and 190,000 households without water and electricity [18]. Therefore, a coordinated restoration that takes the entire IWPN into account would be beneficial. In this regard, the authors in [19] developed an optimization-simulation method for coupled power-water

system restoration. The method optimizes unit commitment and power flow for restoring power, and pump/tank operations to restore water service. Decisions for both subsystems are conducted separately with an interface for information updates. In [20], a centralized restoration method is designed for power–water–gas systems using mixed-integer second-order cone programming. Weighted cost for unmet demands is set as the objective to minimize.

In these references, [14–16,19,20], RMs comprise components operation and flow dispatching. Nevertheless, other RMs, especially network reconfiguration, which are also effective in improving system restoration, should also be exploited for a more resilient restoration scheme. In [21], the spanning tree method is applied to reconfigure a damaged power distribution network (PDN) into islands. It defines root buses that connect distributed generators (DGs) and then links other buses to build radial sub-networks. Besides spanning tree, the single-commodity-flow method is another way to decide on reconfiguration. It generates fictitious networks with identical topology to real networks, makes adjustments to fictitious networks, and then applies them to real ones [22]. Similar to PDN, the reconfigurability of the water distribution network (WDN) should also be exploited. Different from PDN which requires radial operation, WDN are usually looped or meshed, but with strict pressure constraints. For example, a linear programming method is proposed in [23] to reconfigure water networks for better pressure management and zone partitioning. Ref. [24] developed a multi-objective optimization to do the reconfiguration and reduce valve actions during the restoration. These reconfiguration algorithms consider the states of valves individually while neglecting the overall WDN topology and connectivity. Hence, they risk the isolation of some demand nodes, which leads to an unguaranteed minimum water supply for consumers. In addition, the joint reconfigurability of water and power networks is yet to be reported in the literature.

Apart from the joint reconfiguration identified above, handling uncertainties is also a key issue, since they can considerably affect the efficacy and reduce the efficiency of mentioned RMs. Uncertainties exist in renewable generations [25], and water/electricity demands [26]. For most RMs, decisions need to be made before the uncertainties' realizations.

To make reliable decisions under uncertainties, robust optimization (RO), and stochastic programming (SP) are widely accepted. Generally, RO methods focus on the solutions that perform well under the worst-case realization of uncertain variables to guarantee the robustness [27]. For example, it is used to handle the uncertainties of water/power demands and provide reliable coordination of water and power generation [28]. In [29], a RO model is proposed for the planning of energy–water nexus systems under demand uncertainties. Although the algorithm provides robustness technically, a notable limitation of RO methods still exists: their decisions against worst cases are overly conservative because the probability of that worst-case scenario can be extremely low in real-world conditions. Different from RO methods, the SP method is to optimize the objective considering the expectation of uncertainties, which can relieve the conservatism of solutions from RO. In [30], SP was proposed to enhance the IWPNS' resilience, with components hardening and system operation. However, SP can be over-optimistic, as it neglects some outlier situations with low frequency but high impacts. To avoid this, risk-averse SP is proposed to include outlier conditions in risk measures. The authors in [31] studied the IWPNS restoration with risk-neutral and risk-averse SP and show that risk-averse SP works better under outlier scenarios. It is also noted that other methods are also used to handle uncertainties, such as chance-constrained optimization [32], Monte Carlo simulation [33], and reinforcement learning [34]. However, they frequently need to assume probabilities and do not guarantee robustness.

Finally, the above RO and SP methods [28–31] only consider single timescale decisions to determine all the RMs with uncertainties in one batch before executing the decisions together. As the estimation for uncertainties can be inaccurate at an early stage, the decisions can lose

optimality. Moreover, for some RMs with faster response speeds (RSs), their actions can be tuned. That means the operator can make better decisions on these RMs upon receiving more accurate and/or updated information. Thus, a multi-timescale decision framework leveraging these aspects can achieve better restoration by retaining robustness and mitigating over-conservatism. A prototypical example in an unrelated field is in the day-ahead planning for electricity markets [35]. However, to the best of authors' knowledge, such multi-scale algorithms have not been applied for IWPNS' restoration in a continuous manner.

From the above literature review, the research issues for IWPNS restoration can be summarized below:

- (i) Lacking a restoration-oriented and specified IWPNS model that considers interdependent physical features down to component level and consumer behaviour.
- (ii) Insufficient utilization of restoration methods, especially the joint reconfigurability of IWPNS systems.
- (iii) Absence of a multi-timescale framework for continual adaptation to uncertainties and timely tuning of decisions after obtaining more accurate estimates of the uncertainties.

To study these above research issues and demonstrate an improved overall restoration performance of IWPNS systems, this paper proposes a multi-timescale risk-averse restoration framework for IWPNS with component interdependencies modelled, joint reconfigurability formulated, and uncertainties adaptability considered. The framework focuses on IWPNS resilience at the community-scale distribution level since it is close to consumers and various resources. The key contributions are as follows:

- (i) A network-based coordinated restoration model for IWPNS. The proposed model integrates renewable generations, component-wise interdependencies, and couplings in water and power demands. Pipe leakage and storage features are further included in water restoration.
- (ii) More restoration methods (RMs) are utilized, and particularly joint reconfiguration for more effective restoration. All the methods are grouped based on their response speeds (RS) to decision changes to provide flexibility.
- (iii) A comprehensive multi-timescale decision framework to improve the restoration by sequencing the RMs and in the treatment of uncertainties. The framework exploits a timely updated predictions of uncertainties and adjusts the faster RS tunable decisions to improve the IWPNS restoration. The method combines RO and risk-averse SP and is formulated in mixed-integer linear programming (MILP), which can be efficiently solved by commercial solvers.

2. Restoration-oriented IWPNS model

2.1. Structure of IWPNS system

A general structure for IWPNS is illustrated in Fig. 1. It comprises two layers representing power and water supply systems. In the power system layer, there is a radial distribution grid connecting power generation and consumption [25]. The generation part comprises DGs, solar, and wind generations. The consumption side includes energy storage systems (ESSs), and various power appliances. The water system layer contains pipe networks, water components, and consumptions. Involved components are pipelines, valves, and storage tanks. Different from PDN, the topology of WDN can be radial, looped, or even meshed [14]. The interdependencies between water and power subsystems show in components that work for water systems but require power, such as treatment plants and pumps. Treatment plants process raw water and produce potable water. Pumps can be further grouped depending on their location, capacity, and function. That is, pump stations to maintain the pressure of water networks, tank pumps to fill

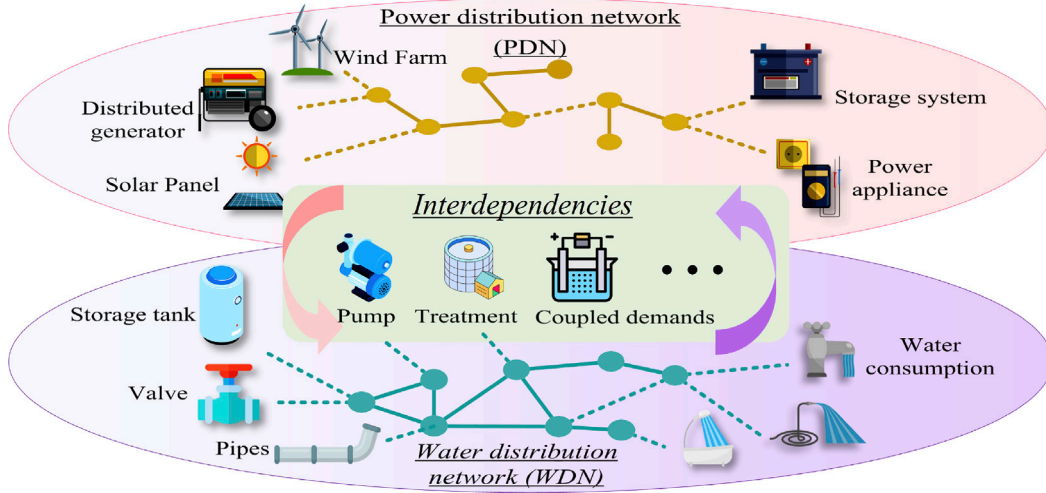


Fig. 1. A general structure for IWPN systems.

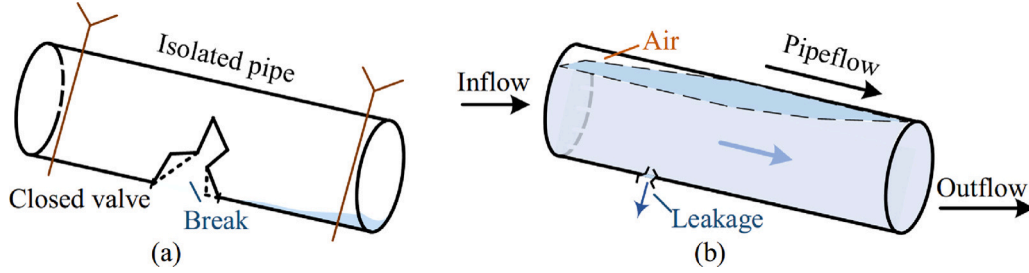


Fig. 2. Illustration for (a) pipe breakage and (b) leakage and storage.

and release water, and water booster pressure systems (WBPS) to pump water for high-rise buildings.

The coupling within IWPN is also shown on the consumer side, which involves residential, industrial, and commercial consumptions. Some industries, e.g., construction, metallurgy, and hydrogen production, further have consumption of water and power nearly keeping to certain ratios [11]. Various residential activities rely on both water and power supplies simultaneously with similar water–power nexus for commercial consumption.

The following sections give detailed description for IWPN components. While existing component models are used, new/modified models developed here include pipe breakage, leakage, and storage model ((1)–(7)), modified head loss model (Eqs. (11)–(12)), and joint network reconfiguration model (Eqs. (44)–(53)).

2.2. WDN model

2.2.1. Pipe breakage and leakage

Pipelines form the WDN and convey the potable water from treatment units to demand nodes. Failures in pipelines can fully or partially interrupt the water transmission, which corresponds to breakage and leakage status [24]. A severe breakage can quickly drain out the pipe and needs to be isolated immediately and removed from the pipeline set. Then the water flow through that pipe equals zero, as given in Eq. (1) and Fig. 2(a).

$$Q_{i,j,t} = 0, \forall (i,j) \notin \mathcal{L}, t \quad (1)$$

On the other hand, light damage can cause water leakage in pipes, as shown in Fig. 2(b). In this case, the leakage flow relates to the size and location of the damage, as well as pressures of both ends [36]. For the ease of quick identification of parameters, it is approximately modelled in Eq. (2) that the water leakage of a pipe is linearly related

to the water head of end nodes and location of the leak. Under emergencies, the leakage flow can drain the pipes and introduce air into pipes. To include this feature, each pipe is modelled as a water storage unit. The equality between water inflow and outflow is relaxed to allow the fill and release of air. Instead of the traditional water flow model that treats water pipes as simple links, Eq. (3) improves the equation to consider water inflow ($Q_{i,j,t}$), outflow ($-Q_{j,i,t}$), leakage ($Q_{i,j,t}^l$) and air replacement ($Q_{i,j,t}^{air}$) for pipes. The accumulation of air replacement is indicated in Eq. (4), where the air in the pipe is accumulated by the airflow rate and time span. Constraint (5) limits the air volume in functioning pipes ($\lambda^W = 1$) for maintaining water transfer. For isolated pipes, constraint (5) is relaxed. Constraint (6) indicates that the leakage should stop when the pipe is drained out, where δ is a small value.

$$Q_{i,j,t}^l = c_{i,j}^{leak} \frac{u_1 \mathcal{H}_{i,t} + u_2 \mathcal{H}_{j,t}}{u_1 + u_2}, \text{ if } \mathcal{V}_{i,j,t}^{air} \leq \mathcal{V}_{i,j} - \delta, \forall (i,j) \in \mathcal{L}, t \quad (2)$$

$$Q_{i,j,t} + Q_{i,j,t}^{air} - Q_{j,i,t}^l = -Q_{j,i,t}, \forall (i,j) \in \mathcal{L}, t \quad (3)$$

$$\mathcal{V}_{i,j,t+1}^{air} = \mathcal{V}_{i,j,t}^{air} + Q_{i,j,t}^{air} \Delta T, \forall (i,j) \in \mathcal{L}, t \quad (4)$$

$$0 \leq \mathcal{V}_{i,j,t}^{air} \leq \mathcal{V}_{i,j,t}^{vol} \mathcal{V}_{i,j} + M^W (1 - \lambda_{i,j}^W), \forall (i,j) \in \mathcal{L}, t \quad (5)$$

$$Q_{i,j,t}^l = 0, \text{ if } \mathcal{V}_{i,j,t}^{air} \geq \mathcal{V}_{i,j} - \delta, \forall (i,j) \in \mathcal{L}, t \quad (6)$$

2.2.2. Pipe storage

Pipe storage is utilized here for improving water resilience. It is to actively extract water from pipes for emergency use by allowing more outflow than inflow with air to fill the space. It represents an additional water source especially when the pipe is disconnected from the WDN after reconfiguration. Constraint (7) limits the flow rate that the system can actively withdraw from pipe storage, which is represented by air inflow rate minus leak rate.

$$Q_{i,j,t}^{air} - Q_{i,j,t}^l \leq r_{i,j}^{lim} Q_{i,j}^{max}, \forall (i,j) \in \mathcal{L}, t \quad (7)$$

2.2.3. Water flow and pressure

For each water node, Eq. (8) represents mass conservation among water from treatments ($\mathbb{Q}_{i,t}^{tr}$), tanks ($\mathbb{Q}_{i,t}^d$, $\mathbb{Q}_{i,t}^c$), pipe flows ($\mathbb{Q}_{i,j,t}$), and restored water service ($\mathbb{W}_{i,t}^r$). Pipeflows are bounded in constraint (9) [37].

$$\mathbb{Q}_{i,t}^{tr} + \mathbb{Q}_{i,t}^d - \mathbb{Q}_{i,t}^c = \sum_{(i,j) \in \mathcal{E}} \mathbb{Q}_{i,j,t} + \mathbb{W}_{i,t}^r, \forall i \in \mathcal{I}, t \quad (8)$$

$$-\mathbb{Q}_{i,j}^{\max} \leq \mathbb{Q}_{i,j,t} \leq \mathbb{Q}_{i,j}^{\max}, \forall (i,j) \in \mathcal{L}, t \quad (9)$$

Eq. (10) is the Hazen–Williams equation that shows water head loss in pipes as given by [38]. It is a power function of water flow while the coefficient is determined by the length (L), roughness (C) and diameter (D) of the pipes.

$$\mathcal{H}_{i,j,t}^{loss} = \frac{\alpha^{hw} L_{i,j}}{(C_{i,j})^{\beta_{hw}} (D_{i,j})^{\gamma_{hw}}} \times (\mathbb{Q}_{i,j,t})^{\beta_{hw}}, \forall (i,j) \in \mathcal{L}, t \quad (10)$$

As per the above description, both leakage flow and pipe storage can accumulate air in pipes and induce higher head loss [39]. This feature is modelled in Eq. (11) as an additional increment for head loss, relating to the ratio of air replacement. Eq. (12) represents the overall head change through a pipe, where the pressurization from pumps ($\mathcal{H}_{i,j,t}^{pu}$) is also included.

$$\mathcal{H}_{i,j,t}^{loss,air} = \left[1 + r_{i,j}^{air} \frac{\mathcal{V}_{i,j,t}^{air}}{\mathcal{V}_{i,j,t}} \right] \mathcal{H}_{i,j,t}^{loss}, \forall (i,j) \in \mathcal{L}, t \quad (11)$$

$$\mathcal{H}_{i,t} - \mathcal{H}_{j,t} = \mathcal{H}_{i,j,t}^{loss,air} - \mathcal{H}_{i,j,t}^{pu}, \forall (i,j) \in \mathcal{L}, t \quad (12)$$

Water pressure is calculated as head minus elevation, as in Eq. (13). Constraint (14) represents pressure-driven analysis for WDN [37], i.e., the water service is limited by percentage when pressure is lower than the designed value, and zero when it is lower than the minimum.

$$\mathcal{P}_{i,t} = \mathcal{H}_{i,t} - \mathcal{E}_{i,t}, \forall i \in \mathcal{I}, t \quad (13)$$

$$\mathbb{W}_{i,t}^r \leq \max \left\{ 0, \frac{\mathcal{P}_{i,t} - \mathcal{P}_{i,t}^{\min}}{\mathcal{P}_{i,t}^{\text{des}} - \mathcal{P}_{i,t}^{\min}} \times \mathbb{W}_{i,t}^r \right\}, \forall i \in \mathcal{I}, t \quad (14)$$

It is noted that besides the Hazen–Williams formula, the Darcy–Weisbach formula is also widely used for the head loss of WDN [40], though to a lesser extent for piped systems. Considering their computation burdens after linearization are similar, we selected the former formula due to a higher accuracy being achieved when linearizing.

2.2.4. Delayable water demand

Constraints (15)–(17) show the delayable water demand feature for water consumption, which means if current water demand is not fully supplied, but is supplemented shortly afterwards, it is still acceptable [26]. Eq. (15) calculates the unmet water demand for each time step and Eq. (16) accumulates the unmet demands in prior Δt steps in a buffer. Then the buffer is used to increase the upper limit of current restoration, as in constraint (17).

$$\Delta \mathbb{W}_{i,t}^{ud} = \mathbb{W}_{i,t} - \mathbb{W}_{i,t}^r, \forall i \in \mathcal{I}, t \quad (15)$$

$$\mathbb{W}_{i,t}^{buf} = \sum_{t'=\Delta t}^t \Delta \mathbb{W}_{i,t'}^{ud}, \forall i \in \mathcal{I}, t \quad (16)$$

$$0 \leq \mathbb{W}_{i,t}^r \leq \mathbb{W}_{i,t} + \epsilon_i^{dl} \mathbb{W}_{i,t}^{buf}, \forall i \in \mathcal{I}, t \quad (17)$$

2.3. PDN model

2.3.1. Power flow and voltage

For the PDN model, since the connectivity of PDN dominates the restoration performance, rather than power losses in the power flows, we choose DistFlow method for power flow modelling due to the simplicity and wide adoption [20,41]. Eqs. (18) and (19) represent active and reactive power balance in generations from renewables (pv ,

w) and DGs (dg), consumptions from consumers (c) and WDN facilities (Y), as well as ESSs ($^{c/d}$) and power flows.

$$P_{i,t}^{dg} + \bar{P}_{i,t}^{w} + \bar{P}_{i,t}^{pv} + P_{i,t}^c - P_{i,t}^d - P_{i,t}^r = \sum_{(i,j) \in \mathcal{L}} P_{i,j,t} + Y_{i,t}^{tr} \mathbb{A}_{i,j}^{tr} + \left(Y_{i,t}^{pc} - Y_{i,t}^{pg} \right) \mathbb{A}_{i,j}^{tk} + Y_{i,t}^{wb} \mathbb{A}_{i,j}^{wb} + Y_{i,t}^{pu} \mathbb{A}_{i,j}^{pu}, \forall i \in \mathcal{I}, t \quad (18)$$

$$Q_{i,t}^{dg} + Q_{i,t}^w + Q_{i,t}^{pv} - Q_{i,t}^r = \sum_{(i,j) \in \mathcal{L}} Q_{i,j,t}, \forall i \in \mathcal{I}, t \quad (19)$$

Constraints (20) to (22) bound active/reactive/apparent power flow and voltage variation, respectively [42]. Restored power and bus voltage are limited in (23).

$$-\lambda_{ij}^P S_{ij}^{\max} \leq P_{i,j,t}, Q_{i,j,t} \leq \lambda_{ij}^P S_{ij}^{\max}, \forall (i,j) \in \mathcal{L}, t \quad (20)$$

$$-\sqrt{2} \lambda_{ij}^P S_{ij}^{\max} \leq P_{i,j,t} + Q_{i,j,t}, P_{i,j,t} - Q_{i,j,t} \leq \sqrt{2} \lambda_{ij}^P S_{ij}^{\max}, \quad \forall (i,j) \in \mathcal{L}, t \quad (21)$$

$$-M^P \left(1 - \lambda_{ij}^P \right) \leq V_{i,t} - V_{j,t} - \frac{r_{ij} P_{i,j,t} + x_{ij} Q_{i,j,t}}{V_0} \leq M^P \left(1 - \lambda_{ij}^P \right), \quad \forall (i,j) \in \mathcal{L}, t \quad (22)$$

$$P_i^{\min}, Q_i^{\min}, V_i^{\min} \leq P_{i,t}, Q_{i,t}, V_{i,t} \leq \bar{P}_{i,t}, \bar{Q}_{i,t}, V_i^{\max}, \forall i \in \mathcal{I}, t \quad (23)$$

2.3.2. DG generation

Eq. (24) formulates the fuel consumption by DGs. $E_{i,t}^{ex}$ is newly allocated fuel described in Eqs. (54) and (55). The generation capacities for DGs are indicated in (25) [43].

$$E_{i,t}^{dg} = E_{i,t-1}^{dg} - P_{i,t}^{dg} \Delta T + E_{i,t}^{ex}, \forall i \in \mathcal{I}, t \quad (24)$$

$$P_i^{dg,\min}, Q_i^{dg,\min} \leq P_{i,t}^{dg}, Q_{i,t}^{dg} \leq P_i^{dg,\max}, Q_i^{dg,\max}, \forall i \in \mathcal{I}, t \quad (25)$$

2.3.3. ESS operation

Eq. (26) indicates the state of charge (SOC) for ESS units, which is changed by charging and discharging power (P^c/P^d). The boundaries of SOC and charging/discharging power are indicated in constraints (27) and (28) [44].

$$SOC_{i,t} = SOC_{i,t-1} - \frac{\Delta T}{E_{i,t}^{ess}} \left[\frac{P_{i,t}^d}{\eta^d} - \eta^c P_{i,t}^c \right], \forall i \in \mathcal{I}, t \quad (26)$$

$$SOC_i^{\min} \leq SOC_{i,t} \leq SOC_i^{\max}, \forall i \in \mathcal{I}, t \quad (27)$$

$$0 \leq P_{i,t}^c, P_{i,t}^d \leq P_i^{c,\max}, P_i^{d,\max}, \forall i \in \mathcal{I}, t \quad (28)$$

2.4. Interdependent features

The interdependent features in IWPNS comprise component-wise linkages and coupling of demands.

2.4.1. Component-wise interdependencies

Electricity-driven components in the water system include treatment plants, water tanks, pump stations, and WBPSs.

(i) Water treatment plants process raw water into potable water, whose power consumption is linearly modelled in Eq. (29) and bounded in constraint (30) [40].

$$Y_{i,t}^{tr} = r_i^{tr} \mathbb{Q}_{i,t}^{tr}, \forall i \in \mathcal{I}, t \quad (29)$$

$$0 \leq \mathbb{Q}_{i,t}^{tr} \leq \mathbb{Q}_{i,t}^{tr,\max}, \forall i \in \mathcal{I}, t \quad (30)$$

(ii) Water storage tanks are equipped with pressure booster pumps to lift water from the pipe network to tanks. Stored water can also be released back and the released water can power the turbine to produce electricity, as modelled in Eqs. (31) and (32) [45]. Eq. (33) updates the volume in tanks and (34) indicates capacities for tanks and consumed/recovered power.

$$\eta_i^{pc} Y_{i,t}^{pc} = \rho g (\mathcal{H}_i^{tk} - \mathcal{H}_{i,t}) \mathbb{Q}_{i,t}^c \mathbb{A}_{i,j}^{tk}, \forall i \in \mathcal{I}, t \quad (31)$$

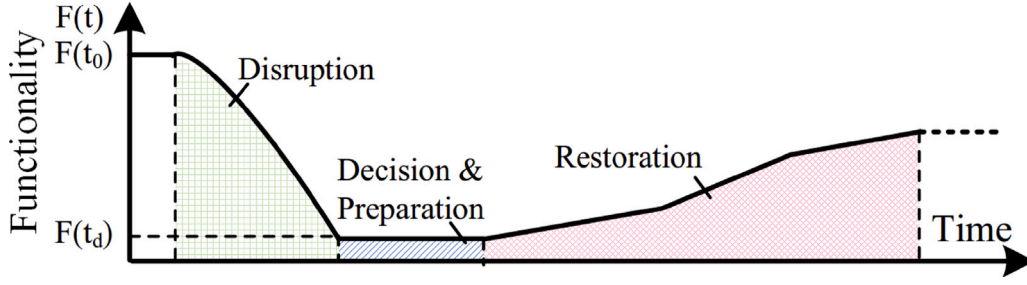


Fig. 3. System functionality curve.

$$\frac{1}{\eta_i^{pg}} Y_{i,t}^{pg} = \rho g (\mathcal{H}_i^{tk} - \mathcal{H}_{i,t}^d) \mathbb{Q}_{i,t}^d \mathbb{A}_{J,i}^{tk}, \forall i \in \mathcal{J}, t \quad (32)$$

$$\mathcal{V}_{i,t}^{tk} = \mathcal{V}_{i,t-1}^{tk} + (\mathbb{Q}_{i,t}^c - \mathbb{Q}_{i,t}^d) \Delta T, \forall i \in \mathcal{J}, t \quad (33)$$

$$0 \leq \mathcal{V}_{i,t}^{tk}, Y_{i,t}^{pg} \leq \mathcal{V}_{i,t}^{tk, \max}, Y_{i,t}^{pg, \max} \quad \forall i \in \mathcal{J}, t \quad (34)$$

(iii) Pump station model contains head gain and Eq. (36) for energy consumption. Both are functions of water flows as formulated in Eqs. (35) and (36), respectively [26]. Eq. (37) limits the power consumption for pump stations.

$$\mathcal{H}_{i,j,t}^{pu} = \sigma^a \mathbb{Q}_{i,j,t} + \sigma^b, \forall (i, j) \in \mathcal{L}, t \quad (35)$$

$$\eta^{pu} Y_{i,j,t}^{pu} = \sigma^c \left(\sigma^a \mathbb{Q}_{i,j,t}^2 + \sigma^b \mathbb{Q}_{i,j,t} \right), \forall (i, j) \in \mathcal{L}, t \quad (36)$$

$$0 \leq Y_{i,j,t}^{pu} \leq Y_{i,j,t}^{pu, \max}, \forall (i, j) \in \mathcal{L}, t \quad (37)$$

(iv) A WBPS is to supply water in a building pipeline with consistent pressure throughout the home [46]. It consists of a pump and a pressure tank and boosts water pressure by actively pumping water into the system. The power consumption of WBPSs is indicated in Eq. (38).

$$Y_{i,t}^{wb} = r_i^{wb} \mathcal{W}_{i,t}^r \quad (38)$$

2.4.2. Coupled power and water demands

The coupling between power and water consumption represents simultaneous demands for both water and power [11]. Shortage of any one of them can hinder the activity, which in turn limits the consumption of the other supply. A set of mutual constraints is applied to model this feature for both restorations, as in (39) and (40).

$$\mathcal{W}_{i,t}^r \tilde{P}_{i,t} \leq \omega_{i,j}^{I,\mathcal{J}} \mathcal{W}_{i,t}^r P_{i,t}^r, \quad \forall (i, j) \in \mathbb{A}_{I,\mathcal{J}}^{nex}, t \quad (39)$$

$$\tilde{W}_{i,t}^r P_{i,t}^r \leq \xi_{i,j}^{I,\mathcal{J}} \mathcal{W}_{i,t}^r \tilde{P}_{i,t}, \quad \forall (i, j) \in \mathbb{A}_{I,\mathcal{J}}^{nex}, t \quad (40)$$

3. Multi-timescale risk-averse resilient operation

3.1. System functionality curve

The system functionality curve of an IWPN before and after disruptions is shown schematically in Fig. 3. The system states usually include the following:

- (1) Disruption occurrence: disruption happens, and system performance drops.
- (2) Decision and preparation: System operators collect information about impacted components and available resources, then make decisions for system restoration.
- (3) Restoration: The operators conduct RMs to restore the system functionality.

3.2. Formulation of the optimization problem

3.2.1. Objective function

The objective (*Obj*) to be minimized is shown in (41), which is a summation of the cost for unmet demands (C^{Loss}) in Eq. (42) and the penalty for switch/valve actions (*Pnt*) in Eq. (43), where $\mathcal{L}^0$ and $\mathcal{L}^0$ are the sets for connected water/power linkages before the reconfiguration. The four terms represent costs to disconnect valves, disconnect switches, connect valves, and connect switches, respectively.

$$Obj = C^{Loss} + Pnt \quad (41)$$

$$C^{Loss} = \Delta T \sum_{i \in \mathcal{I}} \left[\sum_{j \in \mathcal{J}} w_{j,t}^w (\mathcal{W}_{j,t}^w - \mathcal{W}_{j,t}^r) + \sum_{i \in \mathcal{I}} w_{i,t}^p (\tilde{P}_{i,t} - P_{i,t}^r) \right] \quad (42)$$

$$Pnt = c_{ij}^w \sum_{(i,j) \in \mathcal{L}^0} (1 - \lambda_{ij}^W) + c_{ij}^p \sum_{(i,j) \in \mathcal{L}^0} (1 - \lambda_{ij}^P) + c_{ij}^w \sum_{(i,j) \notin \mathcal{L}^0} \lambda_{ij}^W + c_{ij}^p \sum_{(i,j) \notin \mathcal{L}^0} \lambda_{ij}^P \quad (43)$$

The RMs are described in the following section.

3.2.2. Joint reconfiguration

Water network reconfiguration supports restoration by shutting some leaking pipes and connecting backup pipes. It is feasible due to large installations of remotely controllable valves and backup components. The constraints for connectivities are formulated in (44) to (46). Eqs. (44) and (45) ensure consistency for connection states and fictitious flow values. Constraint (46) limits fictitious flow values according to the connection states. In (47), the root water nodes are chosen for the network reconfiguration. Constraint (48) applies the reconfigured topology to the studied water network.

$$\lambda_{i,j}^W = \lambda_{j,i}^W, \quad \forall (i, j) \in \mathcal{L} \quad (44)$$

$$\vartheta_{i,j} = -\vartheta_{j,i}, \quad \forall (i, j) \in \mathcal{L} \quad (45)$$

$$-M^W \lambda_{i,j}^W \leq \vartheta_{i,j} \leq M^W \lambda_{i,j}^W, \quad \forall (i, j) \in \mathcal{L} \quad (46)$$

$$-M^W \mu_i^W \mathcal{F}_i^r \leq \sum_{(i,j) \in \mathcal{L}} \vartheta_{i,j} + 1 \leq M^W \mu_i^W \mathcal{F}_i^r, \quad \forall i \in \mathcal{J} \quad (47)$$

$$\mathbb{Q}_{i,j,t} \leq \lambda_{i,j}^W \mathbb{Q}_{i,j,t}^{max}, \quad \forall (i, j) \in \mathcal{L} \quad (48)$$

According to (48), once the pipe (i, j) is reconfigured to disconnect, the water flow will be non-positive. It means the inflow to the pipe will be shut but outflow is still permitted such that the downstream node can still get water from pipe storage. Physically it is realized by closing the high-pressure side valve and opening the low-pressure side valve.

PDN reconfiguration is to pick up disconnected loads and maintain radial operation. The single commodity flow (SCF) method is applied to determine the new topology [42]. SCF assumes a fictitious network with the same topology as the PDN, with generation nodes as sources, demand nodes as sinks, and with identical fictitious demands. Then, the method constrains the number count of links and satisfaction of demands, thus guaranteeing the radiality and connectivity of reconfigured PDN. In the below, Eq. (49) gives the radiality of the network. Constraints (50) to (51) formulate the conservation of the linkage status

Table 1
Summary for RMs and decision variables.

Response speed (RS)	Restoration measure (RM)	Decision variables
Type I: Low RS Variable set: X_I	(a) Joint reconfiguration (b) Potable water treatment (c) Fuel allocation	$\lambda_{ij}^P, \mu_i^P, \lambda_{i,j}^W, \mu_i^W$ $\mathbb{Q}_{i,j}^{tr}, Y_{i,j}^{tr}$ E_i^{ex}
Type II: Medium RS Set: X_{II}	(d) Tanks and pumps (e) Distributed generations (f) Energy storage dispatch	$\mathbb{Q}_{i,j}^c, \mathbb{Q}_{i,j}^d, \mathcal{H}_{i,j}^{pu}, Y_{i,j}^{wb}$ $P_{i,j}^{dg}, Q_{i,j}^{dg}$ $P_{i,j}^c, P_{i,j}^d, SOC_{i,j}$
Type III: High RS Set: X_{III}	(g) Selective restorations	$P_{i,j}^r, W_{i,j}^r, \Delta W_{i,j}^{ud}, \Delta W_{i,j}^{bdf}$

and fictitious flow. In (52), zero flow is required for disconnected links, Constraint (53) sets the fictitious demand of sink nodes in the network.

$$\sum_{(i,j) \in L} \lambda_{ij}^P = 2 \left(|N| - \sum_{i \in N} \mu_i^P \right), \forall i \in I \quad (49)$$

$$\lambda_{ij}^P - \lambda_{ji}^P = 0, \forall (i, j) \in L \quad (50)$$

$$\theta_{ij} + \theta_{ji} = 0, \forall (i, j) \in L \quad (51)$$

$$-M^P \lambda_{ij}^P \leq \theta_{ij} \leq M^P \lambda_{ij}^P, \forall (i, j) \in L \quad (52)$$

$$-M^P \mu_i^P I_i^{rt} \leq \sum_{(i,j) \in L} \theta_{ij} + 1 \leq M^P \mu_i^P I_i^{rt}, \forall i \in I \quad (53)$$

It is necessary to note that the connectivity constraints for WDN reconfiguration provide essential assurance for demanding nodes to access water. Since the radially requirement is not needed in the WDN, the reconfiguration method in (44)–(48) includes more flexibility.

Compared with no reconfiguration and PDN-only reconfiguration methods, the advantages of joint reconfiguration are (1) it can employ the redundancy of both WDN and PDN; (2) it considers interdependent links for the reconfigured networks and avoids water components losing access to power.

3.2.3. Fuel allocation

Fuel allocation is to distribute stored fuel to DGs. The allocated fuel for each DG is limited by total capacity in (54). Eq. (55) indicates the arrival time of distributed fuel.

$$\sum_{i \in T} \sum_{i \in I} E_{i,t}^{ex} \leq E^{cap} \quad (54)$$

$$E_{i,t}^{ex} = 0, \forall i \in I, t \in T \setminus t_i^{dvr} \quad (55)$$

3.2.4. Components operation and selective restorations

Other RMs are to organize the operations of various components, such as treatment plants, DGs, ESSs, tanks, and pumps. Selective restorations are the decisions to prioritize loads based on their weights ($w_{i,t}^w$ and $w_{i,t}^p$) and keep the generation-consumption balance for the IWP system.

All the RMs and their decision variables are summarized in Table 1. To facilitate the problems' formulation, we define the set of Type I, II, and III decision variables as X_I , X_{II} , and X_{III} , respectively. The set of random variables is ξ .

3.2.5. Deterministic optimization program

Based on the above formulations, we first formulate a deterministic optimization program for the restoration of IWP systems, as shown in (56).

$$\begin{aligned} \min_{X_I, X_{II}, X_{III}} \quad & \text{Obj} \\ \text{s.t.} \quad & \text{WDN: (1)–(17)} \\ & \text{PDN: (18)–(28)} \\ & \text{Interdependence: (29)–(40)} \\ & \text{Objective and RMs: (41)–(55)} \end{aligned} \quad (56)$$

It is noted that (56) assumes no uncertainties in the optimization. The proposed multi-timescale framework for handling the uncertainties is described next.

3.3. Multi-timescale risk-averse framework

The motivation to develop the multi-timescale risk-averse framework as sketched in Fig. 4 is to take advantage of (a) increasing clarity in uncertainties and (b) tunable decisions for RMs. In this, the optimization in (56) is performed via a long-timescale program at time steps of ΔT^L followed by short-timescale (time steps ΔT^S) and real-time optimizations (time steps ΔT^R) taking advantages of RMs with medium and high RSs as listed in Table 1.

In this problem, the uncertainties comprise the power and water demands as well as solar and wind generation. Uncertainty realizations are performed before making a decision, which, traditionally, once made, will not be adjusted anymore. However, the accuracy of such uncertainty predictions tends to decrease with an increasing temporal gap between the prediction and the actual time of execution. This is because the decision-maker can get more updated information on uncertain variables and make more accurate predictions. If such updated predictions are used to adjust RMs, the efficiency of restoration can be considerably improved.

In realistic restoration, some RMs can be timely tuned or changed, but not all. For example, the production of drinking water comprises coagulation, sedimentation, filtration, and disinfection. The overall process, from raw water to produced drinking water, can take a couple of hours. In this context, we define the ability of RMs to respond to the change of decisions as “response speed (RS)”. A lower RS means a longer response time to decision changes, while a higher RS vice versa. As for RM tunability, they are grouped into three types according to different RS, as listed in Table 1.

Based on the above analyses, the proposed multi-timescale restoration framework is to re-execute the decision process of some fast response operations, to take advantage of updated predictions. At the top level, a “long-timescale” RO program is to decide low-speed Type I RMs (a), (b), and (c) using earlier and less accurate predictions of uncertain variables. These decisions are valid for the whole study period, T^L . Then, a sequence of SP optimizations works to determine the medium-speed Type II RMs (d), (e), and (f), with a later and more accurate prediction of uncertainties. The period is selected as T^S with the optimization program running every ΔT^S and updating the decisions. Finally, real-time optimizations are made to determine the Type III RM (g) with high speed. As this decision process is very close to execution, we consider the uncertainties to be clearly defined and the problem is deterministic.

3.3.1. Long-timescale optimization

At this stage, the task is to make decisions for Type I RMs (a), (b), and (c) with low RS. Due to their characteristics that only allow one-time decisions, The decisions are expected to be feasible for all possible realizations of uncertainties. Meanwhile, the uncertainty information at this point comprises only ranges of them, as formulated in (57).

$$\begin{cases} \mathcal{W}_{j,t}^{\min} \leq \mathcal{W}_{j,t} \leq \mathcal{W}_{j,t}^{\max} \\ \hat{P}_{i,t}^{\min} \leq \hat{P}_{i,t} \leq \hat{P}_{i,t}^{\max} \\ \hat{P}_{i,t}^{w,\min} \leq \hat{P}_{i,t}^w \leq \hat{P}_{i,t}^{w,\max} \\ \hat{P}_{i,t}^{pv,\min} \leq \hat{P}_{i,t}^{pv} \leq \hat{P}_{i,t}^{pv,\max} \end{cases}, \forall j \in \mathcal{J}, i \in I, t \quad (57)$$

With these uncertainties, the problem can be represented as an RO program in (58) for the decisions on X_{II} , X_{II} and X_{III} .

$$\begin{aligned} \min_{X_I, X_{II}, X_{III}} \quad & \{\max_{\xi} \text{Obj}^L = \text{Obj}\} \\ \text{s.t.} \quad & \text{WDN: (1)–(17)} \\ & \text{PDN: (18)–(28)} \\ & \text{Interdependence: (29)–(40)} \\ & \text{Objective and RMs: (41)–(55)} \\ & \text{Estimate of uncertainties: (57)} \end{aligned} \quad (58)$$

As mentioned earlier, a multi-timescale decision framework leveraging updated information of uncertainties and tunable RMs can further

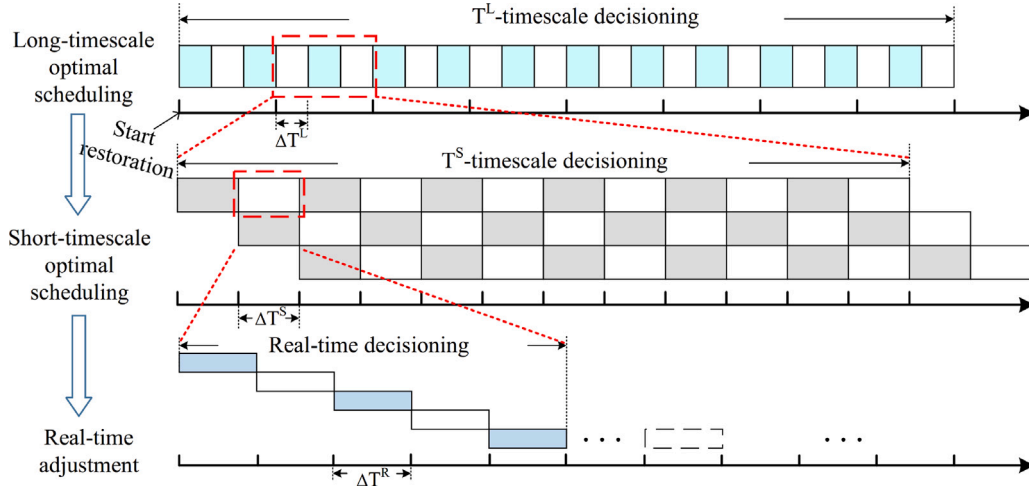


Fig. 4. Structure of multi-timescale optimization.

improve the restoration performance. Note that the long-timescale optimization in (58) computes decision variables X_I , X_{II} and X_{III} . However, only X_I will be applied, and the other decisions will be gradually improved by the following short-timescale and real-time optimizations. In this way, the over-conservation decisions from RO can be gradually mitigated by the following steps, which are discussed next.

3.3.2. Short-timescale optimization

The decisions for the short timescale are to make decisions for Type II RMs (d), (e), and (f). It is conducted by a rolling process with time horizon T^S and step ΔT^S .

The period of concerned uncertainties is much shorter than that in the long-timescale program. Hence, we can obtain more accurate predictions for the random variables, which are represented as probability distributions as in Eq. (59).

$$\begin{cases} f_{\mathcal{W}}(\tilde{\mathcal{W}}_{i,t}) = \mathbb{P}(\mathcal{W}_{i,t} = \tilde{\mathcal{W}}_{i,t}) \\ f_P(\tilde{P}_{i,t}) = \mathbb{P}(P_{i,t} = \tilde{P}_{i,t}) \\ f_{P^w}(\tilde{P}_{i,t}^w) = \mathbb{P}(P_{i,t}^w = \tilde{P}_{i,t}^w) \\ f_{P^{pv}}(\tilde{P}_{i,t}^{pv}) = \mathbb{P}(P_{i,t}^{pv} = \tilde{P}_{i,t}^{pv}) \end{cases}, \forall j \in \mathcal{J}, i \in \mathcal{I}, t \quad (59)$$

To take advantage of such probabilities in Eq. (59), we adjusted the base program in (56) into a risk-averse SP. First, we assign the decisions for the Type I RMs using results from long-timescale optimization. Then, we set Type II RMs as first-stage SP decisions, while the Type III RMs are set as second-stage decisions. In the formulations of the SP program, the first-stage variables remain unchanged for all scenarios of uncertainty realization, and the second-stage variables can adapt to each scenario [25]. Hence, we add a “scenario” dimension (subscript: s) for Type III RMs and related constraints.

The risk-averse optimization objective is set as a composite of conditional value at risk (VaR) and the expectation of uncertain scenarios, as in (60) and (61). For each scenario, C_s^{Loss} is the aggregated cost for unmet water and power demands. η is the mean VaR of solutions for all scenarios and C^η computes the average of $(1-\alpha)\%$ worst-case scenarios with a weighting of ρ .

$$C_s^{Loss} = \Delta T \sum_{i \in \mathcal{T}} \left[\sum_{j \in \mathcal{J}} w_{j,t}^w (\mathcal{W}_{j,t,s} - \mathcal{W}_{j,t,s}^r) + \sum_{i \in \mathcal{N}} w_{i,t}^p (P_{i,t,s} - P_{i,t,s}^r) \right], \forall s \in \mathcal{S} \quad (60)$$

$$C^\eta = \rho \left\{ \eta + \frac{1}{1-\alpha} \mathbb{E} [\max(C_s^{Loss} - \eta, 0)] \right\} \quad (61)$$

The optimization program for the short timescale decisions on X_{II} , X_{III} is summarized as follows:

$$\begin{aligned} \min_{X_{II}} \quad & \text{Obj}^S = \mathbb{E} \{ C_s^{Loss} \} + C^\eta \\ \text{s.t.} \quad & \text{1st stage: (25)–(28), (31)–(38)} \\ & \text{2nd stage: (1)–(23)}_s, (39)–(40)_s, \forall s \in \mathcal{S} \end{aligned} \quad (62)$$

3.3.3. Real-time optimization

The real-time optimization is to decide the load curtailments and power/water flows (high RS) in detail. The time horizon for this stage is T^R , with time step ΔT^R . At this stage, the Type I and II RMs are already determined by previous optimizations. The target for this subproblem is to finalize the decisions for Type III RMs. We consider the uncertainties regarding renewables, and demands are cleared, i.e. negligible. Hence, the real-time decision is a deterministic optimization problem, as shown in (63) on X_{III} .

$$\begin{aligned} \min_{X_{III}} \quad & \text{Obj}^R = C^{Loss} \\ \text{s.t.} \quad & (1)–(23), (39)–(40). \end{aligned} \quad (63)$$

3.4. Solution method

3.4.1. Linearization:

The base model in (56) contain nonlinear constraints: (2), (6), (10), (11), (14), (31), (32) and (36). They are then linearized to form a MILP problem.

In them, constraints (2), (6), and (14) containing judgments (“if”) can be linearized with the big-M method. Taking Eqs. (2) and (6) as an example, the linearization with a binary variable $\mu_{i,j,t}^{eq2,6}$ is shown in (64).

$$\begin{cases} -M^W(1 - \mu_{i,j,t}^{eq2,6}) \leq \mathcal{V}_{i,j,t}^{air} - (\mathcal{V}_{i,j,t} - \delta) \leq M^W \mu_{i,j,t}^{eq2,6} \\ -M^W \mu_{i,j,t}^{eq2,6} \leq \mathcal{Q}_{i,j,t}^l - c_{i,j,t}^{leak} \frac{u_1 \mathcal{H}_{i,t} + u_2 \mathcal{H}_{j,t}}{u_1 + u_2} \leq M^W \mu_{i,j,t}^{eq2,6} \\ -M^W(1 - \mu_{i,j,t}^{eq2,6}) \leq \mathcal{Q}_{i,j,t}^l \leq M^W(1 - \mu_{i,j,t}^{eq2,6}) \end{cases} \quad (64)$$

For constraints (10), (11), (31), and (32), they are for node pressures and water pumps. Node pressure impacts water restoration only when it is lower than the designed pressure (see Eq. (14)). For water pumps, their power consumption is not major compared with power loads in PDN. Hence, it is considered acceptable to linearize these constraints for the benefit of modelling and solving.

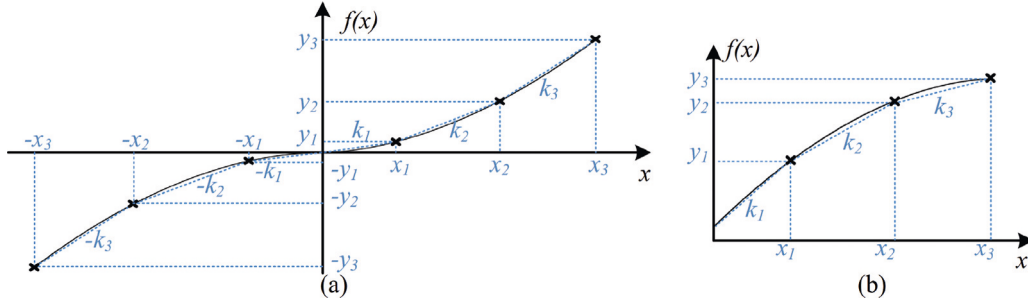


Fig. 5. Linearization for (a) Hazen-Williams equation in (10) and (b) pump power equation in (36). Where x represents $\mathbb{Q}_{i,j,t}$ and $y = f(x)$ is the right-hand side of (10) and (36). $[(x_i, y_i), (x_{i+1}, y_{i+1})]$, in which $i=1,2,3$, are linear segments for representing nonlinear equations. k_i are slopes.

Specifically, we use piece-wise linearization for Eqs. (10) and (36) as shown in Eq. (65).

$$\begin{cases} f^{\text{tar}} = \sum_i (k_i x + \kappa_i y_i) \\ k_i = \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i} \\ \kappa_i x_i \leq x \leq \kappa_i x_{i+1}, \quad \sum_i \kappa_i = 1 \end{cases} \quad (65)$$

where, f^{tar} is the target to linearize and binary κ_i denotes the location of input x . More details for curves and segments are given in Fig. 5.

McCormick envelopes are used to linearize Eqs. (11), (31) and (32). An example for the product $\mathcal{H}_{i,t} \cdot \mathbb{Q}_{i,t}^c$ in (31) denoted by $\Pi_{i,t}$ is shown in (66).

$$\begin{cases} \Pi_{i,t} = \mathcal{H}_{i,t} \cdot \mathbb{Q}_{i,t}^c \\ \Pi_{i,t} \geq \underline{\mathcal{H}}_{i,t} \cdot \underline{\mathbb{Q}}_{i,t}^c + \mathcal{H}_{i,t} \cdot \underline{\mathbb{Q}}_{i,t}^c - \underline{\mathcal{H}}_{i,t} \cdot \underline{\mathbb{Q}}_{i,t}^c \\ \Pi_{i,t} \geq \underline{\mathcal{H}}_{i,t} \cdot \underline{\mathbb{Q}}_{i,t}^c + \mathcal{H}_{i,t} \cdot \underline{\mathbb{Q}}_{i,t}^c - \underline{\mathcal{H}}_{i,t} \cdot \underline{\mathbb{Q}}_{i,t}^c \\ \Pi_{i,t} \leq \underline{\mathcal{H}}_{i,t} \cdot \underline{\mathbb{Q}}_{i,t}^c + \mathcal{H}_{i,t} \cdot \underline{\mathbb{Q}}_{i,t}^c - \underline{\mathcal{H}}_{i,t} \cdot \underline{\mathbb{Q}}_{i,t}^c \\ \Pi_{i,t} \leq \underline{\mathcal{H}}_{i,t} \cdot \underline{\mathbb{Q}}_{i,t}^c + \mathcal{H}_{i,t} \cdot \underline{\mathbb{Q}}_{i,t}^c - \underline{\mathcal{H}}_{i,t} \cdot \underline{\mathbb{Q}}_{i,t}^c \\ \underline{\mathcal{H}}_{i,t}, \underline{\mathbb{Q}}_{i,t}^c \leq \mathcal{H}_{i,t}, \mathbb{Q}_{i,t}^c \leq \overline{\mathcal{H}}_{i,t}, \overline{\mathbb{Q}}_{i,t}^c \end{cases} \quad (66)$$

The reformulated model can be better solved by off-the-shelf solvers.

3.4.2. Robust counterpart for the long-timescale problem

The long-timescale decision in (58) is an RO problem with box uncertainties. After linearization, the problem is feasible and bounded, which meets the requirements for strong duality theory [28]. Hence, it is effective to reformulate the constraints containing random variables into their dual form and eliminate the inner maximization of (58). Then the outcome, the robust counterpart of (58), is a deterministic minimization problem.

3.4.3. Scenario generation/reduction for short-timescale problem

For the scenario-based risk-averse SP problem in (62), the Latin hypercube sampling method is used to generate near-random samples. The simultaneous backward reduction method is then applied to select a smaller but representative set of uncertainties scenarios to facilitate the program, as in Ref. [25].

4. Case study

4.1. Modified 25-node/33-bus case IWPVN

The proposed method is applied on an adjusted 25-node/33-bus community-scale IWPVN case system as shown in Fig. 6.

This case system comprises a WDN layer, a PDN layer [8], and interdependent linkages connecting them. In the WDN, three treatment

Table 2

Pipe leakage coefficients.

Pipe	$c_{i,j}^{\text{leak}}$
1-2, 2-3, 3-4, 4-5, 3-6, 5-6, 6-8, 6-7,	.082, .082, .298, .091, .039, .089,
7-8, 7-9, 9-10, 14-15, 15-17, 17-18,	.081, .089, .048, .019, .041, .018,
15-16, 16-18, 13-14, 13-19, 7-19,	.017, .041, .234, .015, .012, .013,
5-19, 20-1, 21-9, 22-14, 18-23, 23-19,	.006, .187, .012, .051, .235, .171,
13-23, 7-24, 24-13, 11-24, 10-24, 25-6	.251, .071, .067, .095, .002, .055,
	.005

Table 3

Time horizon and step setting.

T^L	ΔT^L	T^S	ΔT^S	ΔT^R
24 h	1 h	3 h	15 min	3 min

plants, three tanks (combined initial water storage of 1600 m³), two pump stations, and several backup pipelines are installed. The pipe network comprises a volume of 4238 m³. Three microgrids (MGs) and one ESS are connected to the PDN. In them, each MG integrates DG sets, solar panels, wind turbines, and local power loads. The remaining fuels for each DG set are 13.04, 13.04, and 15.92 MWh. A depot with 10 MWh fuel is located beside the PDN. The arrival times to MGs are estimated as one, three, and two hours, respectively. For the ESS unit, the rated power and capacity are 100 kW and 400 kWh. Fig. 7 illustrates typical profiles for water demands, power demands, as well as solar, and wind generations. Both WDN and PDN are equipped with switches at branches, and valves at pipelines to support reconfiguration.

After the disruptive event, three power lines in PDN and 11 pipelines were seriously damaged (marked with lighting). They are automatically isolated by relaying protection devices and gate valves. The other power lines are considered safe but the remaining pipes and backup pipes suffered varying degrees of damage, which are modelled by different leakage coefficients, $c_{i,j}^{\text{leak}}$, as estimated in Table 2. The positions of leakages are assumed in the middle of pipes with $u_1 = u_2 = 1$.

As a consequence of disruptions, the behaviour of consumers will be impacted, which leads to uncertain water and power demands. Further, being disconnected from the main grid, the renewables then contribute a considerable amount of available energy resources, and their generation uncertainties necessarily need to be considered. As shown in Fig. 8, water and power demands are estimated by combining recorded data and normal distribution, while solar/wind generations are estimated by Beta and Weibull distributions respectively. It is noted that the clarity of uncertainties experienced is in three stages over the time horizon considered: box uncertainties, probability distributions, and cleared uncertainties. The time horizon and step for each timescale's decision are summarized in Table 3. The weights for water and power loads, i.e., w_j^w and w_j^p are assigned with monetary costs in Ref. [26]. Other parameters' settings can be found in Table 4.

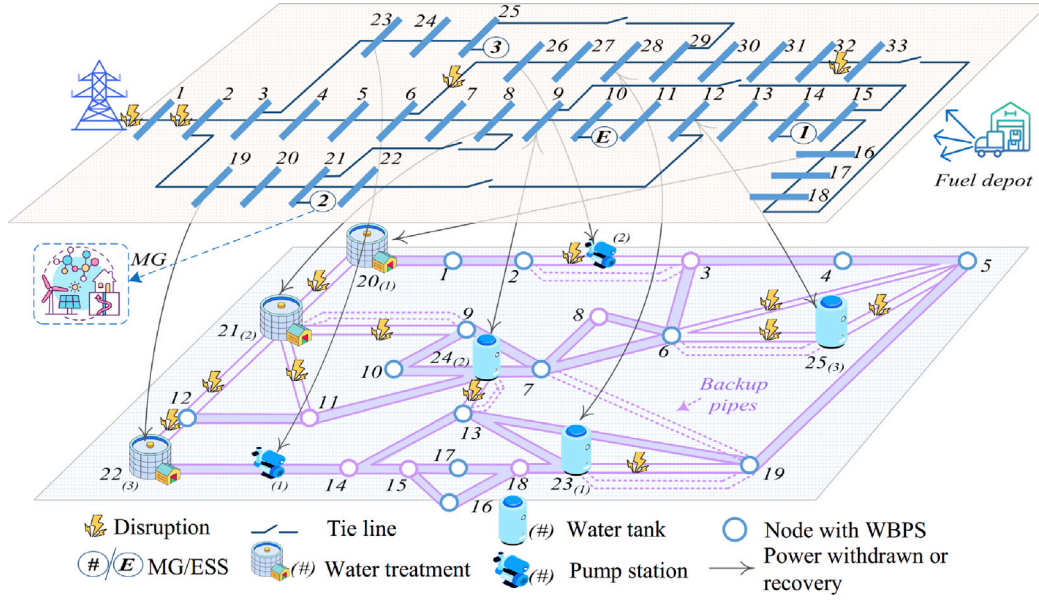


Fig. 6. 25-node/33-bus IWPN case system.

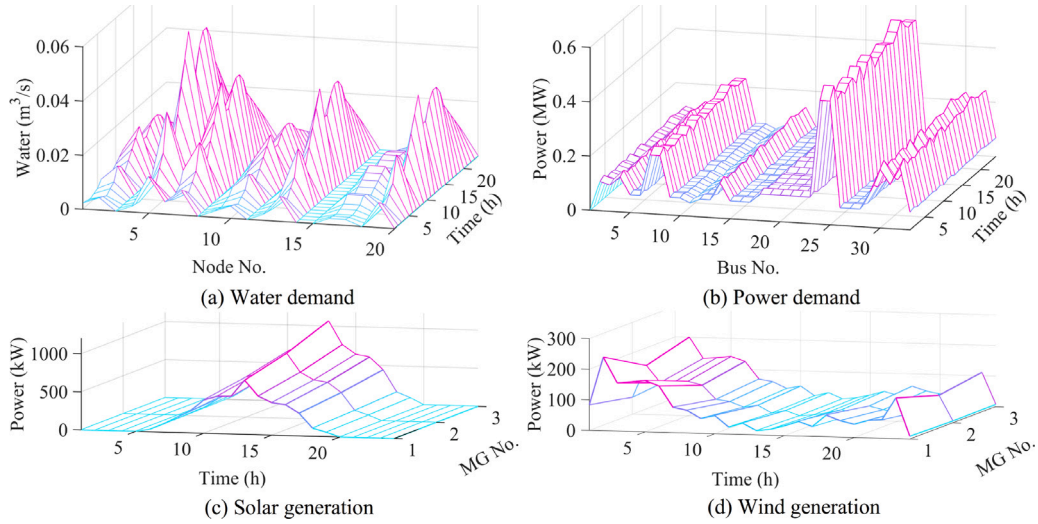


Fig. 7. Profiles of (a) water demand (b) power demand, (c) solar generation, and (d) wind generation.

Table 4

Parameters setting.

$V^{\max/\min}/0$	$P(Q)^{dg,\max}$	$P(Q)^{dg,\min}$	$SOC^{\max/\min}$
1.1/0.9/1.0	1.6MW/1.28MVar	0MW/-12.8Mvar	1/0
r_i^P/r_i^W	M^P/M^W	$\eta_i^0/\eta_i^{\max}$	$p_i^{des}/p_i^{\min}$
1.5e-3 MW/m³	361.1/316.6	600 m³/1200 m³	10 m/0 m
$Q_{i,j}^{\max}$	$r_{i,j}^{lim}/r_{i,j}^{vol}$	$\alpha^{hw}/\beta^{hw}/\gamma^{hw}$	$\varpi_{i,j}^{1-3}/\xi_{i,j}^{1-3}$
0.2 m³/s	0.05/0.05	4.727/1.852/4.871	1.1/1.5

It is noted that other real-world data for IWPNs can also be applied without loss of generality. All the numerical experiments reported below are conducted with Python 3.10 in a 64-bit Windows environment with Gurobi solver (version 10.0.3), on a laptop with AMD Ryzen 7 5800H CPU @3.2 GHz processor and 32 GB of RAM. The parameter MIPGap of Gurobi is set as 0.01%. The solving times for each timescale decision are 212.74 s, 8.40 s (on average), and 0.57s (on average) respectively.

4.2. Numerical results

4.2.1. Long-timescale decisions

During the long timescale decision, Type I RMs with low RS (Table 1) are determined using a min-max RO in (58). Box uncertainties with estimated upper/lower boundaries are used to represent uncertain parameters in Fig. 8. The solutions are executed and effective for the overall time horizon T^L .

RM (a): The decisions for joint reconfiguration are represented in Fig. 9. In Fig. 9(a), WDN is reconfigured by connecting backup pipelines: 6–25, 9–21, 13–24, and 19–23 and disconnecting pipes 6–8, 7–24, 9–10, 13–19, 16–18, and 17–18. It is seen that the algorithm not only chooses the pipes with the largest c^{leak} to isolate but also the redundant pipes. Compared with the original topology, the re-configured WDN contains fewer loops but can still connect to all the demand nodes. Hence, all the nodes are able to access the water source after the reconfiguration. In PDN, bus #1 is isolated as it is unable to connect any sources. Due to the radially constraint, the

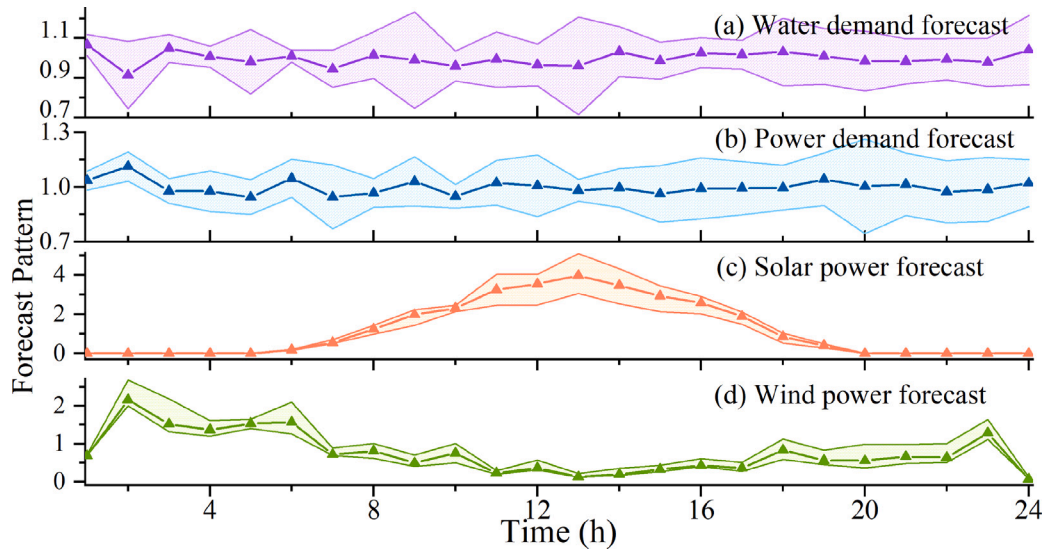


Fig. 8. Forecast for the variations of (a) water demand (b) power demand, (c) solar generation, and (d) wind generation. The lines with triangles are mean values.

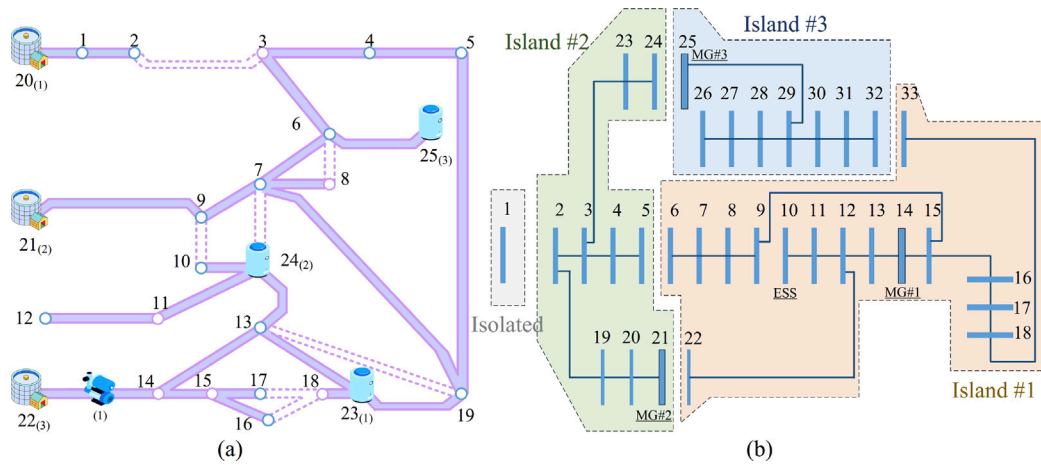


Fig. 9. Long-timescale decision on joint reconfiguration (a) WDN and (b) PDN.

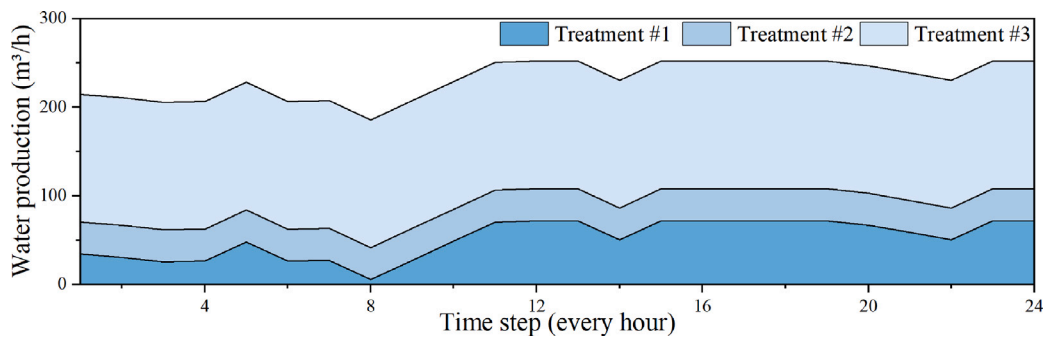


Fig. 10. Long-timescale decision on water production.

other buses are reconfigured into three radial islands and separately supported by three MGs. The joint reconfiguration also considers the interdependent connections. Specifically, three treatment plants are supported by Island #1, the pump station is supplied by Island #2 via bus 23, and three tanks are respectively connected to Island #3 (tank 1) and Island #1 (tanks 2 and 3).

RM (b): The decisions for potable water production are indicated in Fig. 10. With the help of backup pipe 9–21, treatment plant #2

reconnects to WDN. It can be seen that the outputs of treatment plants #2 and #3 reach their capacity, while treatment #1 accounts for varying contribution as it only connects to nodes #1 and #2. The production of water treatment also coordinates with the operation of water storage units (Fig. 11). In the first few hours, the produced water exceeds the demand, as the treatment plants are still at full capacity. The extra water is then stored in water tanks to restore the demands afterwards. It is also noted that the long-timescale decisions

offer strictly robust decisions, which ensures that the water production plan is feasible for any possible realizations of uncertainties bounded in Fig. 8.

RM (c) is to distribute fuels from the depot to three MGs. It can be seen that the assignments are related to the capacity and total demands of each island. Island #1 includes 14 buses, MG #1 and an ESS unit with $SOC = 50\%$. It is assigned 1.95 MWh fuels to complement the power supply and support the operations of connected components in WDN. Island #2 has 9 buses with higher loads, and MG #2, and gets 4.69 MWh of fuel. The third island connects 8 buses and MG #3 with more fuel than the others. It receives 3.36 MWh amount. During the long-timescale decisions, the assignments of fuels are determined together with network reconfiguration. This facilitates the coordination among assignments of loads to islands, dispatch of fuels and regulation of power flow. These decisions are fixed for execution in the whole time horizon T^L . Then the detailed consumption of fuels is determined in the following short-timescale decisions (Fig. 12).

The advantages of applying RO lie in the robustness and feasibility with solutions being valid for any realizations of bounded uncertainties as shown in Fig. 8. In contrast, solutions from non-RO methods can be not possible under the worst-case scenario. For example, if the actual generation of renewables is lower than the prescribed value in the non-RO optimization, the solutions for PDN islanding and fuel assignment can be suboptimal and even infeasible (failed to meet minimum needs). Meanwhile, the conservatism of RO is recognized. Hence, based on always-feasible Type I RMs, the following short-timescale and real-time decisions gradually improve the other RMs' decisions to enhance the restoration.

4.2.2. Short-timescale decisions

After conducting the long-timescale optimizations, the decisions for Type I RMs (a), (b), and (c) are fixed. Then, the short-timescale optimizations perform a rolling process comprising 96 rounds with 15 min time step. At each round, the risk-averse SP in (62) is to determine the Type II RMs (Table 1). With scenario generation and reduction process, ten representative scenarios are selected to represent uncertainties in Fig. 8. The coefficients, ρ , and α are set as 1.0 and 0.8 with reference to [47]. These coefficients are selected as typically used. A further check is via a sensitivity test summarized in Section 4.2.3.

RM (d): Fig. 11 shows the decisions for tank operations. The bar chart indicates water release (positive) and fills (negative), and the curves show the volume change of tank water. From Fig. 11, it can be seen that three tanks get filled at the beginning of the restoration to accommodate the produced water from treatment plants. After that, when the demands grow and are higher than production, they release the stored water into WDN to increase the supplied water. This process also recovers energy to PDN from the geo-potential energy of water. Through the interdependent linkages, the recovered power is transmitted to islands 1 and 3. Hence, the release of tanks can benefit both water and power systems in the peak demand hours. The total volume of water fill/release is 3445 m³.

RM (e) regulates the generations of DGs of each island and corresponding fuel consumption. The results are illustrated in Fig. 12 as bars and curves, respectively. In each round of short-timescale decisions, the probabilities of uncertainties are estimated for the next $T^S = 3$ h with 1515 min time steps. Then, the operations of DGs are optimized to coordinate with renewable generations to supply varying demands in a risk-averse way as in (60) to (62). Thus, the fuels are mainly used in high-demand or low-renewable generation periods. At the end of T^L , three DGs consume the fuels to the lowest level. The total supplied power is 51.31 MWh.

RM (f) is to schedule charging and discharging for the ESS unit. The scheduling coordinates the DG generation and renewables. As shown in Fig. 13, the charging and discharging profiles are illustrated as green bars below and above zero, and the SOC varies accordingly, as the blue curve. After reconfiguration, the ESS unit is organized to cooperate

with MG #1 on Island #1. It absorbs energy during valley periods, and discharges to PDN during peak periods. It also works to smooth the fluctuations of renewable generations. With experiencing alternative charging and discharging processes, it releases all the stored energy the SOC reduces to the lowest and the total energy exchange is 1.64 MWh.

It is worth noting that the short-timescale optimizations are based on the long-timescale decisions. It remains robust and improves the restoration performance with more accurate estimates of uncertainties. Meanwhile, the shortened time step allows more detailed decisions and the rolling process effectively improves the fitness of decisions for the whole period.

4.2.3. Real-time decisions

The real-time decisions contain the dispatch of water/power flows, and selective restoration as shown in (63). At this stage, the uncertainties are fully realized, and the time step for decisions is further shortened to three minutes. Both Type I and II RMs have already been determined by long and short-timescale decisions, such as water and power production as well as components operations. Thus, this stage focuses on detailed flow dispatch and selecting loads to restore.

The load restorations for every node and bus are given in Figs. 14 and 15. At every time step, the limited water and power resources are assigned to demand nodes, according to their accessibility and importance. The importance of loads is measured by the monetary costs. Examples of detailed restoration profiles are given in subplots in Figs. 14 and 15 which expand a 15-time step into details. Finally, 52.3% of the total water demand and 57.3% of the power demand were restored, and the total monetary cost is \$ 345,210.

Fig. 16 shows the trend of air replacement in pipes during the restoration. For the functioning pipes, the air in each pipe is constrained to 5% of the pipe's volume to maintain the transport of water, while the disconnected pipes are drained gradually to support water restoration. The water withdrawn from the pipe storage is 469.68 m³, which accounts for 11% of the total pipe volume.

To check on the sensitivity of parameters α and ρ in the short-timescale SP optimizations, we vary the values of both parameters and keep all other settings constant. This test shows that the best α and ρ are over [0.7, 0.9] and [0.75, 1.0] respectively, with the final objective values in (63) being within 0.23 to 2.46 percent of that using the default setting.

4.3. Performance evaluation

To verify the advantages of the present method (PM), five conventional restoration methods are used for comparison:

(M-A): Independent restoration. The restoration work for WDN and PDN are separately optimized by two independent programs. They assume the other system's status remain in its initial disrupted state. All other settings remain the same.

(M-B): Fixed WDN topology and keeping leaking pipes connected. Operators only conduct the reconfiguration of PDN. All the not-isolated pipes remain connected and backup pipes remain disconnected. All other settings and parameters remain the same.

(M-C): Fixed PDN topology. In this method, only WDN reconfiguration is considered while the PDN topology remains the same. All tie-lines in PDN are kept open.

(M-D/M-E): Single/double timescale decisions. The operators only conduct long or long-short timescale optimizations instead of the multi-timescale optimization. The other settings are kept constant.

The results are shown in Table 5. It can be seen that for M-A, the WDN suffers a huge restoration loss due to the lack of support from the PDN. The PDN saves energy for WDN components but does not supply power to partial customers because of unmet water demands (constraints (39) and (40)). It also loses the flexibility of tanks and pumps. The comparison between M-A and PM showcases the importance of considering interdependency and making restoration plans

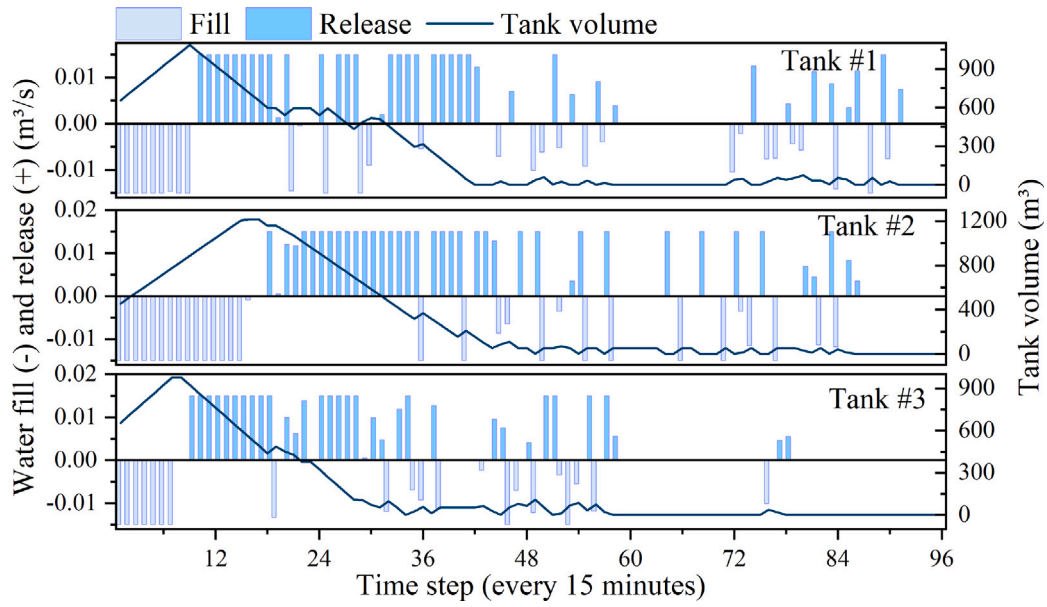


Fig. 11. Short-timescale decision on water fill, release, and tank volume.

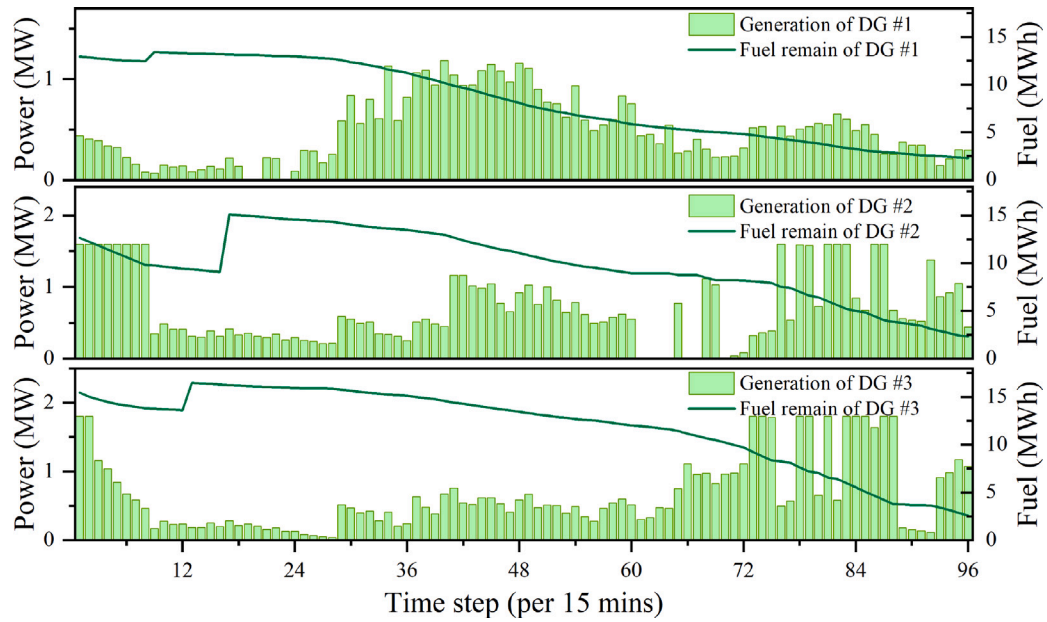


Fig. 12. Short-timescale decision on DG generation and fuel usage.

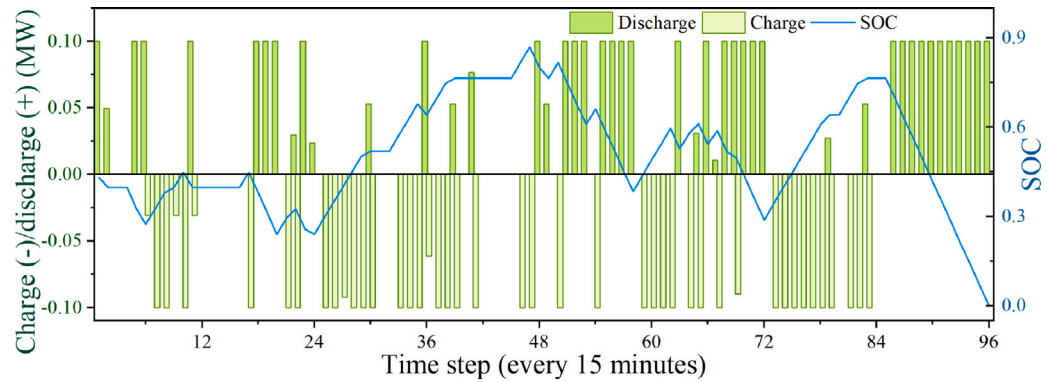


Fig. 13. Short-timescale decision on ESS schedule.

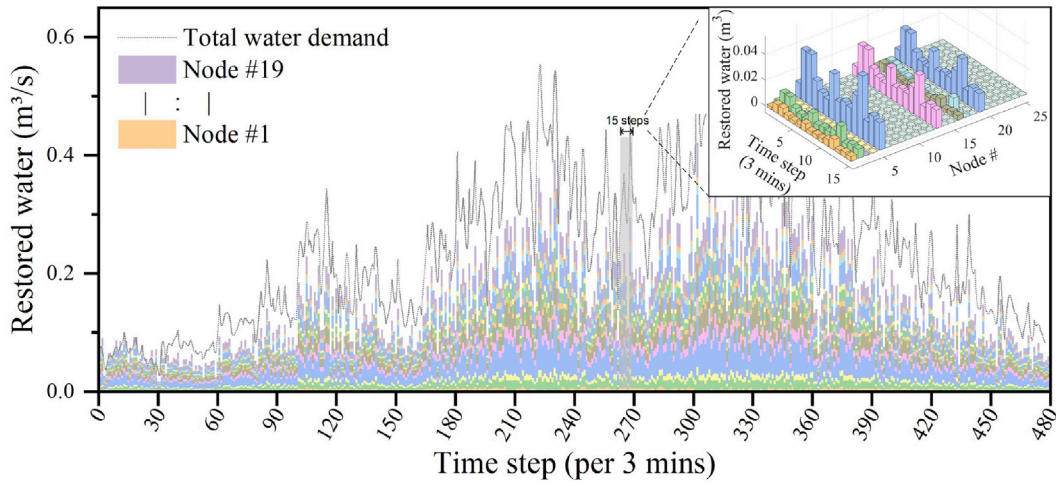


Fig. 14. Realtime decisions on water load restoration (52.3% water load restored at the end of 24 h).

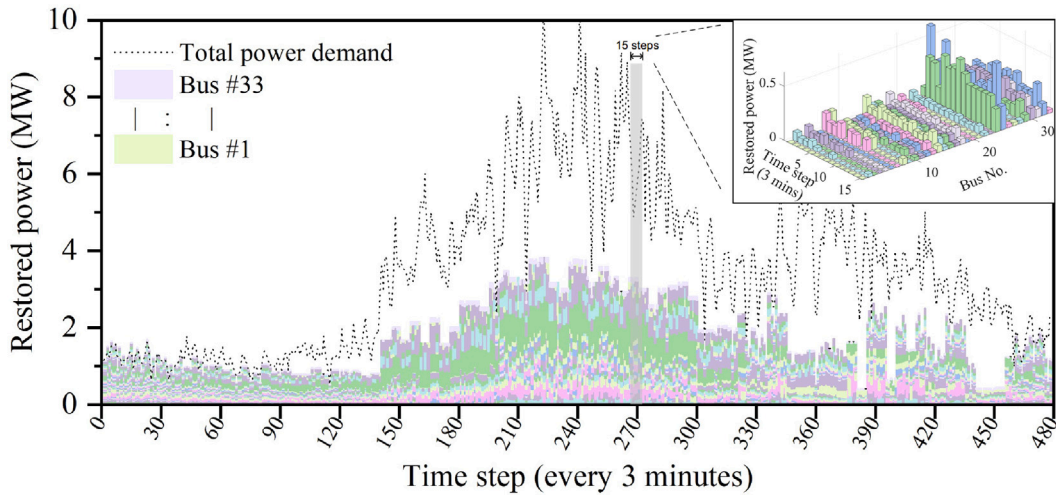


Fig. 15. Real-time decisions on power load restoration (57.2% power load restored at the end of 24 h).

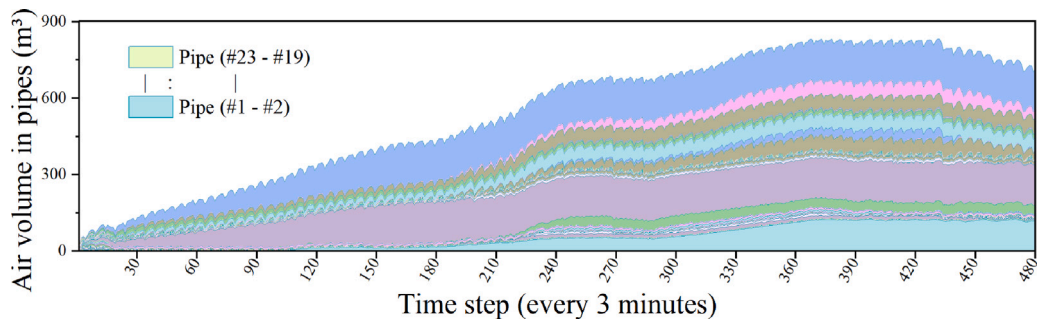


Fig. 16. The volume of air in pipes during the restoration.

together. As the two systems are physically intertwined, neglecting these connections will decrease the restoration of both.

For **M-B**, fixed topology makes WDN unable to mitigate the water leakage and pressure loss caused by damaged pipes. Having the leaking pipes still connected, pipe storage is prohibited from releasing water from them. They both lead to the reduction of overall restoration. Compared with **M-B**, the advantage of doing WDN reconfiguration can be recognized. By actively modifying the on/off status of pipes, WDN can deliver the supplied water with a more efficient path, reduce leakage, and exploit the water of disconnected pipes.

From the comparison of **M-C**, it can be seen that fixed PDN topology considerably hinders the restoration of energy service. The power flow is limited in the surviving lines, but the remaining network fails to connect all buses. With **M-C**, buses #26–#33 are at outage and cannot access any sources. Meanwhile, these unmet power demands also impact WDN components and demands.

The comparison among **M-D**, **M-E**, and **PM** indicates the performance of the multi-timescale framework in achieving better restoration and relieving the conservatism of the RO in (58). **M-D** is to execute all the decisions by (58), including Type II and III RMs. The performance is

Table 5
Methods comparison.

Method	M-A	M-B	M-C	M-D	M-E	PM
Restored water (%)	28.5	35.0	43.4	50.1	51.2	52.3
Restored power (%)	56.5	48.0	41.7	47.2	55.1	57.2

the lowest of the three methods, especially for the PDN with uncertain renewables. **M-E** adds short timescale optimizations in (62), which relieves the conservatism and improves the restoration. The **P-M** applies the whole framework and achieves the best performance.

The above comparisons demonstrate the effectiveness and efficiency of the proposed model and framework with the advantages of including interdependencies, exploiting joint reconfiguration, and better handling uncertainties.

5. Conclusion

In this paper, a multi-timescale restoration method is proposed for the IWP system. A restoration-specified IWP model is developed to integrate features: interdependencies, joint reconfigurability, pipe leakage and storage, and renewable generations to help restore the services. To mitigate the distortion from uncertainties, increasing clarity of uncertain variables and different response speeds of RMs are exploited. A multi-timescale framework is developed to combine RO and risk-averse SP for organizing RMs. Numerical cases are simulated to validate the effectiveness of the method with the following insights:

- (i) Coordinated IWP restoration shows superiority to individual restorations. The interdependencies within IWPNs need critical attention when managing restoration.
- (ii) Joint reconfiguration introduces flexibility to IWPN topology for mitigating leakage and restoring loads. Disconnected pipes provide extra stored water for restoration.
- (iii) The proposed restoration framework can effectively track the clarity of uncertainties and timely adjust the RMs. Compared with traditional methods, it shows more immunity to various uncertainties from water and power demands as well as renewable generations.

The following research can be conducted in the future:

- (1) Distributionally RO with ambiguity sets can be utilized instead of RO to overcome the conservativeness.
- (2) The unbalances in the three phases of PDN can be modelled to improve the IWP model with more physical details.
- (3) Component repair can be added to model the full recovery beyond immediate restoration.

CRedit authorship contribution statement

Yesen Yang: Writing – original draft, Visualization, Software, Methodology, Investigation. **Zhengmao Li:** Writing – review & editing, Supervision, Conceptualization. **Edmond Y. Lo:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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